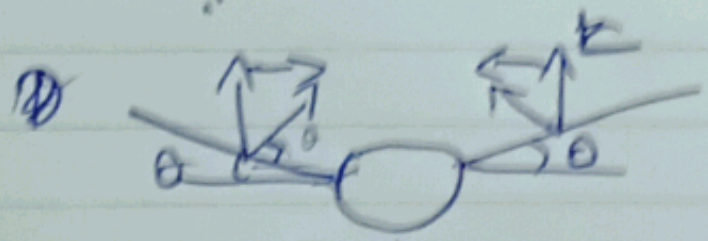
Sweep : ↳ To reduce drag in near and sonic regions
↳ Reduces effect of lift

↳ The flow is not normal
↳ So sweep is not suggested for STOLs

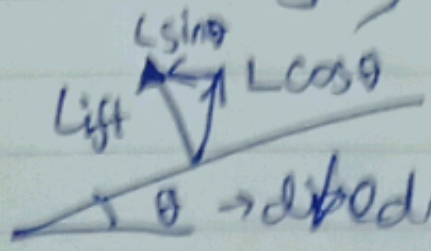
Dihedral and Anhedral

For every 10° sweep we need 1° increase in Dihedral angle



Front view

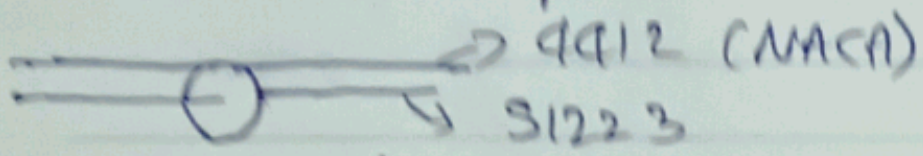
On a wing ;



the total y component is $L\cos(\theta)$ with dihedral

The horizontals simply cancels out.

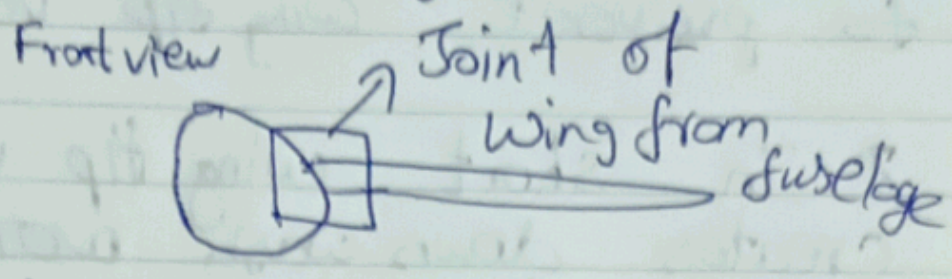
So if, we care about stability of aircraft we can use high wing system with no dihedral or sweep



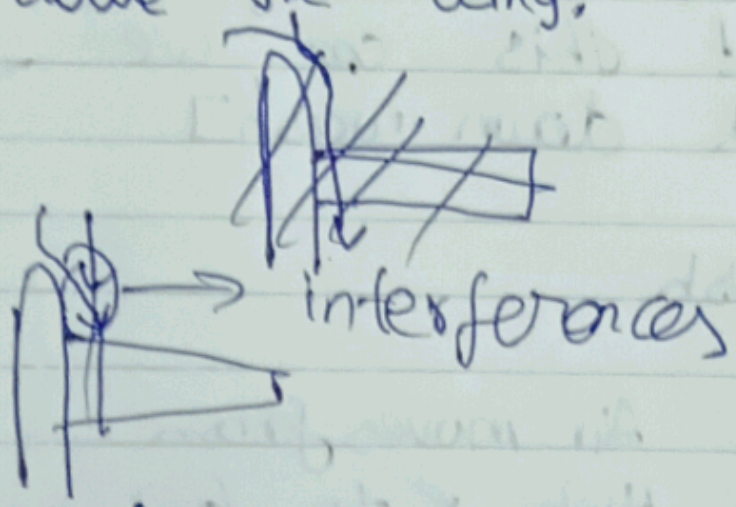
advantages,

- ↳ It is equalant to adding 5° dihedral [stability]
- ↳ The wing is normal to airstream ~~and so n~~ [No side slips]
- ↳ fuselage doesn't affect flow over the wing
- ↳ As we are using multi elemental wing ~~now~~. placement of those elements is not a big constraint now
- ↳ The net Lift is not decomposed now
- ↳ Reverts Interference Drag

Interference Drag :-

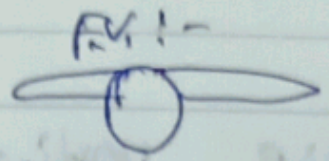


Air flowing around the fuselage interferes the one flowing above the wing.



But if wing is placed on top

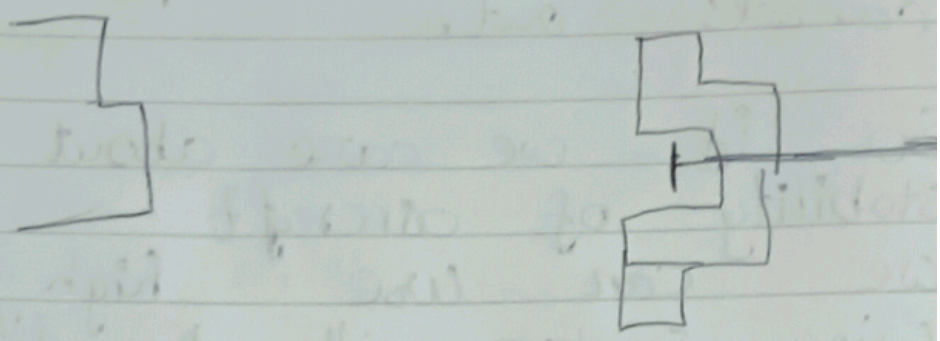
Ex :-



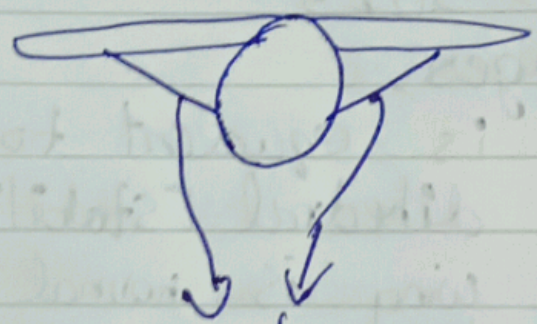
No interference in joints

So to sum up, Sweep - unnecessary [we are in subsonic]

Dihedral → replaced by
↓
High wing system



Disadvantages :-
Structure :-



we have to decide on this

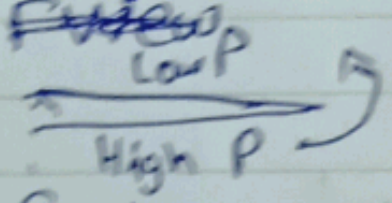
Winglets :-

These are necessary to prevent wing tip vortices

* In short wing tip vortices creates down wash over the ~~sale~~ wing

* [I am still confused about this coz we read down wash]

Down wash



Air moves from High P to low P causing vortices

these vortices are pushed back due to ~~fa~~ upstream air and these vortices push the air downwards

[Need opinions] on this